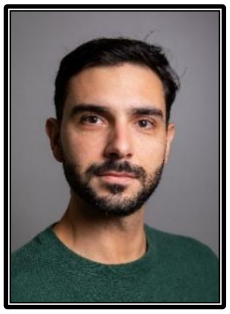
Stefano **Racioppi**, Ph.D.

✉ sr2200@cam.ac.uk | Google Scholar | LinkedIn | orcid.org/0000-0002-4174-1732 | Home Page



Appointments

01/2027 Assistant Research Professor, *University of Cambridge*, UK.

Professional Experience

09/2025 – 12/2026 Research Associate, *University of Cambridge*, UK.
Group: Prof. Chris Pickard.

03/2021 – 08/2025 Postdoctoral researcher, *State University of New York at Buffalo*, USA.
Group: Prof. Eva Zurek.

07/2023 – 09/2023 Visitor, *Aalto University*, Finland.
Group: Prof. Patrick Rinke.

02/2019 – 02/2021 Postdoctoral researcher, *Chalmers University of Technology*, Sweden.
Group: Prof. Martin Rahm.

Education

10/2015 – 01/2019 Ph.D. in Industrial Chemistry, *University of Milan*, Italy.
10/2016 – 03/2018 Visiting student at the *University of Bern*, Switzerland.
10/2013 – 10/2015 Master's degree in industrial chemistry, *University of Padua*, Italy.
10/2010 – 09/2013 Bachelor's degree in industrial chemistry, *University of Padua*, Italy.

Research Interests

Computational and Theoretical Chemistry – High Pressure Chemistry – Crystal Structure Prediction – Quantum Crystallography – Chemical Bonding – Materials and Planetary Science – Superconductivity – Machine-Learning.

Distinctions and Awards

07/2026 Associate Fellow of the Higher Education Academy (AFHEA).
02/2026 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, Seal of Excellence 2026.
01/2026 Research Development Fellows Program – University of Cambridge.
03/2025 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, Seal of Excellence 2025.
09/2019 Best Ph.D. Thesis Award by the Italian Crystallographic Association (AIC).
07/2017 Outstanding poster prize offered by Wiley – Journal of Computational Chemistry and International Journal of Quantum Chemistry. Workshop on ab Initio Modeling in Solid State Chemistry with CRYSTAL, *Minneapolis*, USA.

Fundings, Proposals and Scholarships

09/2026 **Awarded**: ERC Starting Grant; PI: Dr. Racioppi S.; € 1,500,000.
05/2025 **Awarded**: 1 shot-day (8 shots in total) at Omega Laser Facility at the Laboratory for Laser Energetics (LLE). Project: *Structural Determination of Al_2O_3 at super-Earth Conditions – AlO Discovery*. PI: Dr. Racioppi S.
04/2025 **Awarded**: Beamtime (18 shifts) at the European Synchrotron Radiation Facility (ESRF). Project: *Diagnosing Electride Formation in Ca up to 40 GPa with XANES and SCXRD.* PI: Dr. Chin, D. A.; co-PI: Dr. Racioppi, S.
04/2024 **Awarded**: 34th European Crystallographic Meeting Scholarship; € 300.
09/2023 **Contributed**: *High Energy Density Quantum Matter - Renewal*; Award number: LOI-0000043230; Source of Support: Department of Energy (DOE); PI: Dr. Collins G.W.; Co-PIs: Dr. Deemyad S.; Dr. Hemley R. J.; Dr. Zurek E.; $ 2,100,000.
02/2023 **Awarded**: Beamtime at Laser-lab Europe LENS; *Tuning Electrical Conductivity of Aurophilic Interactions by Compression* (ID: 22396); PIs: Dr. Racioppi S., Dr. Poręba T.; Prof. Macchi P.
09/2022 **Awarded**: *Seed Funding Project* – Center for Matter at Atomic Pressure (CMAP); $ 3500.
10/2018 **Awarded**: Scholarship; National Interuniversity Consortium of Materials Science and Technology (INSTM); PI: Prof. Sironi A.; € 5862.
05/2018 **Awarded**: Travelling Grant; Ettore Majorana Foundation; € 600.

Publications

H-index: 11

(33) Liu, Z.; **Racioppi, S**.; Hilleke, K. P.; Pandit, A.; Ma, S.; Hermann, A.; Yan, D.; Zurek, E.; Miao, M. Non-local Chemistry Driven by Cation-Anion Size Disparity in Helium Inserted Compounds under High Pressure, *arXiv:2511.23400*.

(32) **Racioppi, S.***; Zurek, E. Quantum Effects or Theoretical Artifacts? A Computational Reanalysis of Hydrogen at High-Pressure, *arXiv:2510.02098.*

(31) Ocampo, I. K.; Kim, D.; Smith, R. F.; Coppari, F.; **Racioppi, S.**; Zurek, E.; Rygg, J. R.; Chin, D. A.; Millot, M.; Eggert, J. H.; Duffy, T. S.; Atomic-level structure and pressure-density response of Fe-O compounds at deep Earth and exoplanetary interior conditions, *under review* in *Physics of the Earth and Planetary Interiors*.

(30) Dutta, R.; Lazicki, A.; Han, S. K.; Ocampo, I. K.; Smith, R. F.; Gorman, M. G.; Singh, S.; **Racioppi, S.**; Koizumi, S.; Hamel, S.; Braun, D.; Das, P.; Zurek, E.; Krygier, A.; Eggert, J. H.; Duffy, T. S. Synthesis of a Novel Eight-Coordinated Phase of Mg_2SiO_4 at Pressures of Deep Exoplanetary Interiors, *Nat. Comm.*, **2026**, *accepted*

(29) Farraj, H.; **Racioppi, S.**; Garbarino, G.; Ahart, M.; Mondal, A.; Parra, S. G.; Smith, J. S.; Cohen, R. E.; Zurek, E.; Cabana, J.; Hemley, R. Novel Pressure-Induced Transformations of $PbTiO_3$, *Applied Physics Letters*, **2026**, *accepted*.

(28) Halbert, C.; Lamichhane, A.; Wang, B.; **Racioppi, S.**; Salke, N. P.; Kumar, R.; Marathamkottil, A. H. M.; Farraj, H.; Wang, K.; Poldi, E. H. T.; Ahart, M.; Liu, Z.; Schulze, D. J.; Song, S.; Shi, X.; Ren, Z.; Meng, Y.; Konstantin, G.; Reis, R. D.; Tartaglia, R.; Deng, L. Z.; Chu, C. W.; Zurek, E.; Hemley, R. J. High-pressure structures of $Bi_{0.5}Sb_{1.5}Te_3$ at room temperature and in its superconducting states, *Physical Review Materials*, **2026**, *10*, 094804.

(27) **Racioppi, S.**; Kelly, A.; Zurek, E. Computational Investigation of Al_2O_3 Structures at Super-Earth Interior Conditions, *ACS Earth Space Chem*., **2026**, *10*, 2003-2009.

(26) Weadock, N.; Holle, W.; Myford, K.; **Racioppi, S.**; Ghimire, A.; Ladd, D.; Park, C.; Zurek, E.; Toney, M.; Bridges, F.; Deemyad, S. Pressure-driven helium insertion for structural stability of $CH_3NH_3PbBr_3$ hybrid perovskites, *Chem. Mater.*, **2026**, *38*, 7, 3439-3448.

(25) Caussé, M.; Bozier, K.; Cooke, P. I. C., **Racioppi, S.**; Pickard C. J. Metastability and high-T_c superconductivity in A15-type ternary hydride $YSbH_6$ at moderate pressure, *PRB*, **2026**, *113*, 094521.

(24) Coleman, A. L.; Singh, S.; Lockard, T.; Ocampo, I. K. Lazicki, A. E.; Gorman, M. G.; **Racioppi, S.**; Krygier, A. G.; Wehrenberg, C. E.; Ahmad, R.; Hamel, S.; Han, S.; Ginnane, M. K.; Swift, D. C.; Bonev, S. A.; Zurek, E.; Duffy, T. S.; Eggert, J. H.; McNaney, J.; Smith, R. F. Body-centered cubic phase transformation in gold at TPa pressures, *Physical Review Letters*, **2025**, *135*, 18, 186101.

(23) **Racioppi, S.***; Zurek, E. Using Topology to Predict Electrides in the Solid, *J. Phys. Chem. A*, **2025**, *129*, 10031-10038.

(22) **Racioppi, S.***; Zurek, E. Activation of Semicore Electrons in Alkali Metals and Their Role in the B1–B2 Phase Transition under Pressure, *J. Am. Chem. Soc*, **2025**, *147*, 36, 32745-32751.

(21) Storm, C. V.; **Racioppi, S.**; Duff, M. J.; McHardy, J. D.; Zurek E.; McMahon, M. I. Experimental Evidence of Interstitial Electron Density in Transparent Dense Sodium, *Commun. Mater*, **2025**, *6*, 201.

(20) **Racioppi, S.**; Zurek, E. High-Pressure Electrides: A Quantum Chemical Perspective, *Annu. Rev. Mater. Res.*, **2025**, *55*, 421-442.

(19) **Racioppi, S.**; Saffarin-Deemyad, I.; Holle, W.; Belli, F.; Ferry, R.; Kenney-Benson, C.; Smith, C. J., Zurek, E.; Deemyad, S. Phase Lithium's low-temperature phase transitions: Insights into quantum lattice dynamics and superconductivity, *Physical Review B*, **2025**, *111*, 054111.

(18) **Racioppi, S.**; Zurek, E. Looking at High-Pressure Electrides Through the Lens of Quantum Crystallography: The Case of Simple Cubic Calcium, *Acta Cryst.*, **2025**, B81.

(17) **Racioppi, S.**; De la Roza, A.; Hajinazar, S.; Zurek, E. Powder X-Ray Diffraction Assisted Evolutionary Algorithm for Crystal Structure Prediction, *Digit. Discov.*, **2025**, 4, 73-83.

(16) **Racioppi, S.**; De la Roza, A.; Hajinazar, S.; Zurek, E. PXRD-Assisted Crystal Structure Predictions, *Acta Cryst.*, **2024**, A80, e396.

(15) **Racioppi, S.**; P.; Hyldgaard, P.; Rahm, M. Quantifying Atomic Volume, Partial Charge and Electronegativity in Condensed Phases, *J. Phys. Chem. C*, **2024**, 128, 4009-4017.

(14) **Racioppi, S.***; Storm, C. V.; McMahon, M. I.; Zurek, E. On the Electride Nature of Na-hP4, *Angew. Chem. Int. Ed.* **2023**, e202310802.

(13) **Racioppi, S.**; Miao, M.; Zurek, E. Intercalating Helium into A-site Vacant Perovskites, *Chem. Mater.* **2023**, 35, 4297-4310.

(12) Fang, M.; Srikanth Kumar, G.; **Racioppi, S.**; Zhang, H.; Zurek, E.; Lin, Q. Hydrazonyl Sultones as Stable Tautomers of Highly Reactive Nitrile Imines for Fast Bioorthogonal Ligation Reaction, *J. Am. Chem. Soc.* **2023**, 145, 18, 9959-9964.

(11) **Racioppi, S.**; Lolur, P.; Hyldgaard, P.; Rahm, M. A Density Functional Theory for the Average Electron Energy, *J. Chem. Theory Comput.* **2023**, 19, 799-807.

(10) Antle, J.; Kimura, M.; **Racioppi, S.**; Lang, M.; Damon, C.; Gatley-Montross, C.; Sánchez B., L.; Miller, D.; Zurek, E.; Brown, A.; Gast, K.; Simpson, S. Applying Density Functional Theory to Common Organic Mechanisms: A Computational Exercise, *J. Chem. Educ.* **2023**, 100, 355-360.

(9) **Racioppi, S.**; Orian, L.; Gennaro, A.; Isse, A. Solvent Coordination Effect on Copper-Based Molecular Catalysts for Controlled Radical Polymerization, *Catalysts* **2022**, 12, 1656.

(8) Poręba, T.; **Racioppi, S.***; Garbarino, G.; Morgenroth, W.; Mezouar, M. Investigating the Structural Symmetrization of CsI_3 at High Pressures through Combined X-ray Diffraction Experiments and Theoretical Analysis, *Inorg. Chem.* **2022**, 61, 10977-10985.

(7) Della Pergola, R.; Garlaschelli, L.; Macchi, P.; Ruffo, R.; **Racioppi, S.**; Sironi, A. From Small Metal Clusters to Molecular Nanoarchitectures with Core-Shell Structure: Synthesis, Redox Fingerprint, Theoretical Analysis and Solid-State Structure of $[Co_{38}As_{12}(CO)_{50}]^{4-}$, *Inorg. Chem.* **2022**, 61, 9888-9896.

(6) Kumar, G. S.; **Racioppi, S.**; Zurek, E.; Lin, Q. Superfast Tetrazole−BCN Cycloaddition Reaction for Bioorthogonal Protein Labeling on Live Cells, *J. Am. Chem. Soc.* **2022**, 144, 57–62.

(5) **Racioppi, S.**; Rahm M. In-Situ Electronegativity and the Bridging of Chemical Bonding Concepts, *Chem. Eur. J.* **2021**, 27, 18156-18167.

(4) **Racioppi, S.***; Sironi, A.; Macchi, P. On Generalized Partition Methods for Interaction Energies, *Phys. Chem. Chem. Phys.* **2020**, 24291-24298.

(3) **Racioppi, S.**; Andrzejewski, M.; Colombo, V.; Sironi, A.; Macchi, P. Different Metallophilic Attitudes Revealed by Compression, *Inorg. Chem.* **2020**, 59, 2223−2227.

(2) **Racioppi, S.***; Della Pergola, R.; Colombo V.; Sironi, A.; Macchi, P. Electron Density Analysis of Metal Clusters with Semi-Interstitial Main Group Atoms. Chemical Bonding in $[Co_6X(CO)_{16}]^-$ Species, *J. Phys. Chem. A* **2018**, 122, 5004-5015.

(1) Della Pergola, R.; Sironi, A.; Colombo, V.; Garlaschelli, L.; **Racioppi, S.**; Sironi, A.; Macchi, P. Periodical trends in $[Co_6E(CO)_{16}]^-$ clusters: Structural, synthetic and energy changes produced by substitution of P with As, *J. Organomet. Chem.* **2017**, 849-850, 130-136.

* *Corresponding author*

Talks

04/2027 **Invited**. American Physical Society (APS) Global Physics Summit; Computational Physics.

08/2026 **Invited**. 27th Congress and General Assembly of the International Union of Crystallography; Microsymposium MS084: Quantum crystallography of non-equilibrium or nanostructured materials.

06/2025 **Contributed**. 24th Biennial International Conference on the Science of Compression in Condensed Matter (SCCM): Computational Methods: Theory, Codes and Applications (FMDA/B).

06/2025 **Contributed**. Crystal Structure Prediction Workshop, a Practical Approach, Poitiers.

09/2024 **Contributed**. 4th European Symposium on Chemical Bonding (CBOND2024).

08/2024 **Contributed**. 34th European Crystallographic Meeting (ECM34); Microsymposium M27: High Pressure Crystallography: Exploring Structure and Method Development at Extreme Conditions.

07/2024 **Invited**. Gordon Research Seminar (GRS), Research at High Pressure.

07/2023 **Contributed**. 28th AIRAPT and 60th EHPRG International Conference on High Pressure Science and Technology; Session: Chemical Bonding.

03/2022 **Contributed**. American Physical Society (APS) March Meeting; Session G24: Matter at Extreme Conditions: Phase Transitions.

09/2021 **Invited**. Italian Crystallographic Association (AIC) XLIX Meeting.

06/2018 **Contributed**. International School of Crystallography in Erice; 52nd Course: Quantum Crystallography.

Seminars

10/2026 *Centre for Science at Extreme Conditions (CSEC) Seminar*, University of Edinburgh, *Host*: Dr. Dominique Laniel.

04/2026 *Diamond's Science Happy Hour*, Diamond Light Source Ltd; *Host*: Dr. Annette Kleppe.

03/2026 Italian Young Crystallographers (GCI) *GCInsights* Virtual Seminar; *Host*: Dr. Martina Lippi.

11/2025 Chicago/DOE Alliance Center (CDA) Virtual Seminar Series; *Host*: Prof. Russell Hemley.

05/2025 Cambridge Crystallographic Data Centre (CCDC), Virtual Seminar; *Host*: Dr. Fabio Montisci.

07/2024 High Energy Density Science Center (HEDS) Virtual Seminar Series; Lawrence Livermore National Laboratory (LLNL, USA); *Host*: Dr. Federica Coppari.

01/2024 University of Milan (Italy), Dep. of Chemistry; *Host*: Prof. Davide Proserpio.
08/2023 Aalto University (Finland), Dep. of Physics; *Host*: Prof. Patrick Rinke.
06/2023 University of Oviedo (Spain), Dep. of Chemistry; *Host*: Prof. José Manuel Recio.
04/2023 Italian Young Crystallographers (GCI) *Happy Hour* Virtual Seminar; *Host*: Dr. Marta Morana.
03/2022 Chicago/DOE Alliance Center (CDA) Virtual Seminar Series; *Host*: Prof. Russell Hemley.

Services

07/2026 Chair of the Gordon Research Seminar (GRS), Research at High Pressure, held in Holderness, USA.
07/2026 Co-organizer and chair of the "*Under Pressure: from Chemical Bonding to Planetary Structures and Evolution"* session at the Goldschmidt 2026 conference.
03/2026 Appointed Springer Nature Reviewing Editor of Nature Portfolio journals.
07/2025 Co-organizer of the "*High Energy Density Physics: Quantum Materials and Quantum Phenomena*" session at the 24th Biennial Conference on Science of Compression in Condensed Matter (SCCM).
06/2025 Co-organizer of the workshop on the crystal structure prediction code XtalOpt at the Crystal Structure Prediction Workshop, a Practical Approach, held in Poitiers, France.
06/2022 Co-organizer of the workshop on the crystal structure prediction code XtalOpt at the International School of Crystallography in Erice; 56th Course: Crystallography under extreme conditions.
06/2022 Co-organizer of the workshop on computational chemistry techniques at the International School of Crystallography in Erice; 56th Course: Crystallography under extreme conditions.

Teaching

2025 – 2026 Demonstrator for a course in the Scientific Computing MPhil on Electronic Structure and Density Functional Theory – University of Cambridge.
2024 Special virtual lecture for the general chemistry course at the University of Milan – "*History and Evolution of the Electronegativity*"
2022 – 2023 Online workshops for the usage of the crystal structure prediction code XtalOpt.
2018 – 2019 Teaching assistant in physical chemistry II for 2nd year degree in Industrial Chemistry – University of Milan.
2015 – 2016 Teaching assistant in general and inorganic chemistry for 3rd year degree in Physics – University of Milan.

Supervision

2026 Supervisor of one student for the MPhil project in Scientific Computing (Josh Cheung) and of two students for the MPhil written assignment in Scientific Computing (Rongze Sun, Jiawei Li) – University of Cambridge.
2021 – 2025 Supervisor of two undergraduate students (Amelia Kelly, Jarrod DuMond) – SUNY Buffalo.
Co-supervisor of three PhD students (Morgan Reddington, Masashi Kimura, Isaac Ampersand-Kwadwo Akwetey) – SUNY Buffalo.
2021 Supervisor of a high school student (Henry Feng) for summer project – SUNY Buffalo.
2017 – 2019 Co-supervisor of two master students (Stephanie Terruzzi and Simona Sorbara) – University of Milan.

Memberships and Affiliations

2026 – present Lennard-Jones Centre.
2026 – present Geochemical Society.
2026 – present European Committee for Chemical Bonding.
2025 – present Associate at the Trinity Hall College of Cambridge.
2021 – present American Physical Society (APS).
2018 – present European Crystallographic Association (ECA).
2018 – 2026 Italian Crystallographic Association (AIC).

Computational Developments and Codes

Electronegativity in gas-phase: https://github.com/SteRacioppi/Chi-bar-Atomic-and-Molecular-Electronegativity
Electronegativity in solid: https://www.scm.com/doc/BAND/Expert_Options/Restarts.html

Public Service and Outreach

2026 *Cambridge Festival* – outreach for general public.
2026 *Knowable Magazine*, **Meet the mysterious electrides.**

2025 *News Center - News & information from UB*, ***Core electron bonding may not always require extreme pressure, study finds***.

2025 *Science News*, ***A new iron compound hints 'primordial' helium hides in Earth's core***.

2023 *News Center - News & information from UB*, ***Sodium's high-pressure transformation can tell us about the interiors of stars, planets***.